A-A-53389
14 March 1988

COMMERCIAL ITEM DESCRIPTION

STONE, ARTIFICIAL, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification U-S-746D.

Salient Characteristics:

This commercial item description covers artificial dental stones suitable for making dental stone casts and dies. The stones shall be furnished in the following types and nominal sizes, as specified:

Type I. - High strength stone

Size 1. - 50 grams
Size 2. - 4 1/2 pounds
Size 3. - 35 pounds

Type II. - Low strength stone

Size 3. - 35 pounds
Size 4. - 25 pounds

Type I, high strength artificial stone powder, when mixed with water according to manufacturer's instructions, shall be suitable for fabrication of master casts and dies.

Type II, low strength artificial stone powder, when mixed with water according to manufacturer's instructions, shall be suitable for fabrication, flasking and mounting of diagnostic casts.

Composition and physical properties. The artificial stone shall comply with American Dental Association (ADA), Specification No. 25 (Dental Gypsum Products) for composition and physical properties. Low strength stone shall comply with Type III of ADA Specification No 25. High strength stone shall comply with Type IV of ADA Specification No. 25. Test procedures for setting time, fineness, setting expansion, compressive strength, reproduction of detail, and testing consistency shall be as specified in ADA Specification No. 25.

Color. High strength stone powder shall be furnished in any pale, pastel shade of color, except buff or yellow. Low strength stone powder shall be furnished in a buff or light yellow color only.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Net weight. The stone powder shall be furnished in the weights specified below, when tested using a balance or scale suitable for the purpose intended.

Net Weight of Stone.

Type I	Type II
Size 1. 49-71 grams	Size 3. 35-37 pounds
Size 2. 4 1/2-5 pounds	Size 4. 25-27 pounds
Size 3. 35-37 pounds	

Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the dental stone they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 25 for dental gypsum products. Types I and II of this specification shall be certified for types IV and III, respectively, of ADA specification No. 25. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the stone to the date of delivery to the Government.

Instructions. Instructions shall be as required by ADA specification No. 25 and include a statement that heavy ingredients settle to the bottom, and that for consistently accurate casts, the stone should be mixed thoroughly before dispensing. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes.

Workmanship. The stone shall be free from defects which detract from its appearance or may impair its serviceability.

Unit. Package (PG). One package for size 1, Type I containing twenty-four moisture-resistant envelopes (50-70 grams of stone per envelope) of Type I stone, as specified, constitutes one unit. Each package shall contain instructions for use and description of physical properties.

Unit. Can (CN). One can for size 3, Type I or Type II containing 35 pounds of stone packaged in a recloseable moisture resistant container, as specified, constitutes one unit. Each package shall contain detailed instructions for use.

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Unit. Can (CN). One can for size 2, Type I containing four and one-half pounds of stone packaged in a recloseable moisture resistant container, as specified, constitutes one unit. Each package shall contain instructions for use.

Unit. Can (CN). One can for size 4, Type II containing 25 pounds of stone packaged in a recloseable moisture resistant container, as specified, constitutes one unit. Each package shall contain detailed instructions for use.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified below.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 and 2.5 (percent defective) and an inspection level of II. Unit of product for examination purposes shall be the unit of issue.

Sampling for tests. Sampling for tests shall be conducted on each lot or control number and as specified herein. If any test determination for a lot does not comply with the requirements specified, the lot shall be rejected.

Tests.

Characteristic	Number of determinations per lot or batch number
Consistency	3 1/
Setting time	5 1/
Setting expansion	2 1/
Compressive strength	5 2/
Fineness	2 1/
Net weight	3 3/
Reproduction of detail	2

1/ Each determination shall be the average of two specimens.
2/ Each determination shall be the average of five specimens.
3/ In process testing may be used for this characteristic.

Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

Examination. The stone shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed below.

Defects.

Stone not free from foreign matter, dirt, or impurities.
Stone not free from caked or set lumps.
Instructions for use inadequate, incomplete, inaccurate, or missing altogether.
Color of stone not as specified.
Packaging not as specified.
Packing not as specified.
Marking incorrect, incomplete, illegible, or omitted.

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In process examination may be used in examination for these defects.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder.

In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Type	Size
6520-00-116-1472	STONE, ARTIFICIAL, DENTAL, 35 lb (15.88 kg)	II	3 (CN)
6520-00-557-7000	STONE, ARTIFICIAL, DENTAL, 4 1/2 lb (2.04 kg)	I	2 (CN)
6520-00-557-7015	STONE, ARTIFICIAL, DENTAL, 35 lb (15.88 kg)	I	3 (CN)
6520-00-935-1009	STONE, ARTIFICIAL, DENTAL, 50 Gram, 24s	I	1 (PG)

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project No. 6520-2058

Location: NCR Disk #4

05436

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53389	STONE, ARTIFICIAL, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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